

llmXive Automated Review of IndusAgent: Reinforcing Open-Vocabulary Industrial Anomaly Detection with Agentic Tools

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the factual accuracy of claims and the validity of citations within the manuscript.

Citation Validity and Model Existence: A significant portion of the experimental claims relies on comparisons with models that appear to be non-existent or misnamed. Specifically, the paper frequently references “Qwen3-VL-8B” (e.g., Table 2, Section 4.2, Figure captions) and “GPT-4.1” (Table 1). As of the current public knowledge cutoff and the paper’s own timeline (2025/2026), “Qwen3-VL” and “GPT-4.1” are not standard, publicly released model versions. The authors likely intend to refer to “Qwen2.5-VL” (which is cited elsewhere in the text) or a specific internal variant. Citing non-existent models as baselines undermines the validity of the performance comparisons (e.g., the claim that IndusAgent outperforms “GPT-4.1” by 5.9% on MVTec). The authors must verify the exact model names and versions used in their experiments and correct the citations to reflect actual, verifiable public models or clearly label them as proprietary/internal if that is the case.

Temporal Consistency of References: The bibliography contains numerous citations dated 2026 (e.g., `zhang2026pelican`, `song2026human`, `wang2026cac`, `ADE_CoT`). While the paper itself is an arXiv preprint from 2026, citing works with future publication dates as established literature creates a logical inconsistency. If these are indeed preprints, they should be cited as such with their correct arXiv IDs and dates (e.g., “arXiv preprint 2025” or “in preparation”). Presenting them as 2026 conference/journal publications when the paper is currently a preprint suggests a potential fabrication of publication status or a confusion in the reference management. This affects the credibility of the “Related Work” section.

Reproducibility Claims vs. Checklist: In the NeurIPS Checklist, the authors answer “No” to “Open access to data and code,” justifying it with “data and code will be released upon the publication of our paper.” However, the paper is already available as a preprint on arXiv. While some venues allow code release upon final acceptance, stating “No” for a preprint that is already public can be misleading to readers attempting to reproduce the results immediately. The justification should be more precise (e.g., “Code will be released upon final acceptance to the conference”) or the answer should reflect the current availability if a repository link exists.

Statistical Claims: The paper claims “massive recall surges” (e.g., +17.4% on MPDD). While the numbers in Table 2 support the arithmetic difference, the lack of statistical significance testing (as admitted in the checklist: “doesn’t require error bars”) is a methodological choice that should be explicitly defended in the text rather than assumed. The claim of “SOTA” is supported by the numbers provided, but the validity of the comparison hinges entirely on the accuracy of the baseline model names and versions identified above.

In summary, the core scientific claims are numerically consistent within the provided tables,

but the factual accuracy of the cited baselines and the temporal validity of the references require immediate correction to ensure the paper’s claims are supportable by real-world evidence.

Action items.

- **[writing]** The paper cites ‘Qwen3-VL-8B’ (e.g., Table 2, Fig. captions) and ‘GPT-4.1’ (Table 1) as baselines. These model versions do not exist in public records as of the paper’s context (2025/2026). Verify if these are typos for Qwen2.5-VL or internal unreleased models, and correct citations to match actual public versions or clarify their status.
- **[writing]** The bibliography lists ‘arXiv:2605.20682’ (the paper itself) and several 2026-dated citations (e.g., ‘zhang2026pelican’, ‘song2026human’, ‘wang2026cac’) as established works. Since the paper is a 2026 preprint, citing future-dated works as completed literature is factually inconsistent. Ensure all cited works are either published or clearly marked as ‘in preparation’ or ‘arXiv preprint’ with correct dates.
- **[writing]** The ‘NeurIPS Paper Checklist’ states ‘Answer: No’ for ‘Open access to data and code’ with justification ‘data and code will be released upon publication’. However, the paper claims to be a preprint on arXiv (2605.20682). If the paper is already public, the code/data should ideally be available or the justification should reflect the specific embargo policy. Clarify the availability status to avoid misleading readers about reproducibility.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

The figure effectively illustrates the three reasoning paradigms with clear visual workflows and text examples. However, the caption is truncated at the end, and there is a minor spacing typo in the text within panel (b).

Figure 2

Figure 2 effectively visualizes the three-stage IndusAgent architecture described in the caption. The diagram is clear, with distinct sections for CoT Construction, Fine-Tuning, and RL, and includes necessary details like model names, reward components, and reasoning examples.

Figure 3

The figure effectively demonstrates the qualitative difference between the baseline and the proposed method, but the caption is too generic to stand alone, and the inclusion of a decorative emoji reduces the scientific rigor of the presentation.

Figure 4

The figure effectively demonstrates the qualitative difference between the baseline and the proposed method on a screw defect task, but the caption is too generic to stand alone, and the inclusion of an emoji reduces the scientific presentation quality.

Figure 5

The figure effectively demonstrates the qualitative difference between the baseline and the proposed method, but the caption is too generic to stand alone, and the inclusion of a cartoon emoji undermines the scientific presentation.

Action items.

- **[writing]** Figure 1: The caption text is truncated at the end ('...overcomes both visual ambiguities and physic'), cutting off the sentence and likely the figure reference tag.
- **[writing]** Figure 1: The text in panel (b) contains a formatting error with a missing space between the question mark and the instruction ('...this image ?structure your response...').
- **[writing]** Figure 3: The caption is generic ('Case Study between Qwen3-VL-8B and our method') and fails to describe the specific visual content (cable cross-section) or the specific failure mode (hallucination of normalcy) shown in the image.
- **[writing]** Figure 3: The image contains a decorative crying emoji in the Qwen3-VL-8B panel which is unprofessional for a scientific publication and lacks a definition in the caption.
- **[writing]** Figure 4: The caption is generic ('Case Study between Qwen3-VL-8B and our method') and fails to describe the specific visual content (a screw defect detection task) or the specific outcome shown, making it impossible to interpret the figure's scientific contribution without reading the internal text.
- **[writing]** Figure 4: The image contains a large, distracting emoji (crying face) in the Qwen3-VL-8B panel that is unprofessional for a scientific publication and serves no analytical purpose.
- **[writing]** Figure 5: The caption is generic ('Case Study between Qwen3-VL-8B and our method') and fails to describe the specific visual content (a defect detection task on a textured surface) or the specific outcome shown, making it impossible to interpret the figure without reading the internal text boxes.
- **[writing]** Figure 5: The image contains a large, distracting crying emoji icon in the Qwen3-VL-8B panel; while likely intended to indicate failure, this informal graphic is unprofessional for a scientific publication and should be replaced with a standard indicator or removed.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript exhibits a high density of domain-specific acronyms and jargon that are not consistently defined upon first use, potentially excluding readers from adjacent fields or those new to agentic AI.

In Section 3.2, the term “IoU” is used to describe the spatial localization reward component without definition. While standard in computer vision, a general reader may not immediately grasp its meaning. Similarly, in Section 4.3, the ablation study refers to “TOL” (Tool Augmentation library) without ever spelling out the acronym or defining it in the text. This forces the reader to guess the meaning or search for it elsewhere.

The term “SOTA” is used repeatedly in Section 4.2 and the Conclusion to denote “state-of-the-art.” This is a common shorthand but should be written out in full at the first instance to maintain formal clarity. Furthermore, “CoT” (Chain of Thought) appears in the Related Work and Method sections without an explicit definition, assuming the reader is already versed in reasoning paradigms.

Finally, the acronym “MLLMs” is used in Section 4.2 and Section 5 to refer to “multimodal large language models.” Given the paper’s focus on these models, the full term should be introduced before the acronym is adopted. These instances of undefined jargon create unnecessary friction for non-specialist readers and should be addressed to improve accessibility.

Action items.

- **[writing]** The manuscript exhibits a high density of domain-specific acronyms and jargon that are not consistently defined upon first use, potentially excluding readers from adjacent fields or those new to agentic AI. In Section 3.2, the term “IoU” is used to describe the spatial localization reward component without definition. While standard in computer vision, a general reader may not immediately grasp its meaning. Similarly, in Section 4.3, the ablation study refers to “TOL” (Tool Augmentation library)

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_logical_consistency.md`

“The logical consistency of the paper is compromised by contradictions between the stated reward mechanisms and their mathematical formulation, as well as gaps in the comparative evidence supporting the ‘SOTA’ claim.

First, the **Reward Formulation** (Section 3.2, Eq. 3) contains a critical internal contradiction. The authors define an ‘Accuracy-Gated’ reward where $R_{textacc}$ acts as a multiplicative gate for localization, type, and tool utility. The text explicitly states: ‘If $R_{textacc} = 0$, no credit is given for localization, type, or tools.’ However, the equation is written as $R(\tau) = R_{textacc}(\tau) \cdot (R_{textformat}(\tau) + R_{textloc}(\tau))$.

$R(\tau) = R_{textacc}(\tau)$

$\cdot$

$(R_{textformat}(\tau) + R_{textloc}(\tau))$

$R_{textformat}(\tau)$

Here, $R_{textformat}$ is added *outside* the multiplicative gate. This means an agent that fails the diagnostic ($R_{textacc} = 0$) still receives a positive reward for ‘Process Compliance’ (formatting). This directly contradicts the stated mechanism that the reward should be gated by accuracy. If the goal is to prevent hallucination or incorrect tool use, the formatting reward should also be gated or the logic must be clarified. As written, the agent is incentivized to produce well-formatted but incorrect diagnoses.

Second, the **Main Results** (Section 4.2, Table 1) do not fully support the conclusion that IndusAgent ‘outperforms leading commercial systems’ in a logically rigorous way. The table lists ‘GPT-4.1’ (81.9% Avg) and ‘GPT-4o’ (65.8% Avg). While IndusAgent (83.4%) beats GPT-4.1, the text generalizes this to ‘leading commercial systems’ without clarifying if GPT-4.1 is the specific state-of-the-art commercial baseline or if other unlisted models were tested. Furthermore, the table omits specialized non-agentic industrial anomaly detection methods (e.g., PatchCore, DRAEM) which are standard baselines in the field (cited in Related Work). The claim of ‘SOTA’ is logically weak if it only compares against general-purpose LLMs and ignores the specialized SOTA methods that the paper claims to improve upon. The conclusion that it ‘overcomes perceptual dilution’ relies on the assumption that passive MLLMs are the only alternative, which is not supported by the provided comparison table.

Third, the **Ablation Studies** (Section 4.3, Table 3) present a causal ambiguity regarding the ‘SFT’ component. The text claims omitting SFT causes ‘catastrophic collapse.’ However, the ablation ‘w/o. SFT’ (c) is compared against the full model (b). If SFT is a prerequisite for the RL training (as implied by the ‘anchoring structural consistency’ text), removing SFT likely results in a model that was never properly initialized for the RL phase. The performance drop (76.8% to 57.6%) may reflect the failure of the RL training process itself rather than the specific contribution of the SFT data to the final policy. The experiment does not isolate the effect of SFT on the *inference* capability of a pre-trained model; it conflates the stability of the training pipeline with the utility of the SFT data.

Finally, the **Limitations** section admits to ‘inference overhead’ and ‘dependency on tool feedback reliability,’ which logically undermines the claim of ‘autonomous’ inspection if the tools are unreliable. The paper does not provide a logical argument for how the system handles the case where the tool feedback is noisy (Limitation 2) while simultaneously claiming the reward mechanism is robust. If the tool feedback is noisy, $R_{texttool}$ and $R_{textloc}$ become noisy, potentially leading to the agent learning incorrect policies, yet the paper presents the results as definitive SOTA without discussing the variance or failure modes of the tool feedback loop.”

Action items.

- **[science]** Table 1 claims SOTA but omits specialized baselines like PatchCore. The text generalizes ‘leading commercial systems’ despite only listing GPT-4.1, creating a logical gap in the comparative evidence.
- **[science]** Eq. 3 defines an ‘Accuracy-Gated’ reward but adds `R_format` outside the gate. This allows credit for formatting even when accuracy is zero, contradicting the stated mechanism that failure voids all other rewards.
- **[science]** Ablation ‘w/o SFT’ conflates SFT utility with RL training stability. If SFT is a prerequisite for RL, removing it breaks the pipeline, making the performance drop a training artifact rather than proof of SFT’s specific value.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_overreach.md`

The manuscript makes several claims that extend beyond the immediate evidence provided in the experimental section, specifically regarding the scope of its superiority and the nature of its improvements.

First, the assertion in Section 4.2 and the Conclusion that IndusAgent establishes a new “SOTA” is an overreach. While the model outperforms the specific baselines listed in Table 1 (including commercial APIs and open-source VLMs), the bibliography explicitly cites `miao2025agentiad` (AgentIAD), a specialized agentic industrial anomaly detection method. The absence of AgentIAD from the quantitative comparison in Table 1 suggests the “SOTA” claim is premature or incomplete. If AgentIAD was excluded for specific reasons (e.g., code unavailability), this must be stated, and the “SOTA” claim should be qualified to “among the evaluated baselines.”

Second, the Conclusion states the method “overcomes perceptual dilution and hallucinations.” This is a strong, absolute claim. The experimental results in Section 4.2 demonstrate significant gains in Anomaly Recall (e.g., +17.4% on MPDD), which addresses false negatives. However, the paper does not provide a dedicated metric or analysis quantifying “hallucination” rates (e.g., false positives on normal samples or incorrect defect descriptions) compared to a baseline. Without explicit data showing a reduction in hallucinations, the claim that these are “overcome” is an extrapolation. The language should be tempered to “mitigates” or supported by a specific hallucination analysis.

Finally, the description of “massive recall surges” addressing the “false-negative bottleneck of passive MLLMs” (Section 4.2) relies on a comparison primarily against Qwen3-VL-8B. While this shows the value of the agentic approach over a raw VLM, it does not necessarily prove the method solves the bottleneck relative to the broader field of passive IAD methods (like PatchCore or WinCLIP) which are not evaluated on this specific recall metric in the same

table. The authors should clarify that the “bottleneck” being addressed is specifically that of *general-purpose* MLLMs, rather than implying a universal solution to false negatives in the entire IAD domain.

Action items.

- **[science]** The claim of ‘SOTA’ performance (Section 4.2) is overreaching given the exclusion of recent specialized IAD agents (e.g., AgentIAD, cited in bibliography but not compared in Table 1). The authors must either include these baselines or qualify the claim to ‘among evaluated models’.
- **[writing]** The conclusion states the method ‘overcomes perceptual dilution and hallucinations’ (Section 5). This is an absolute claim not fully supported by the data, which shows high recall but does not explicitly quantify hallucination rates against a baseline. Temper this to ‘mitigates’ or provide specific hallucination metrics.
- **[writing]** The paper claims to address the ‘false-negative bottleneck’ with ‘massive recall surges’ (Section 4.2). However, the comparison baseline for this specific claim (Qwen3-VL-8B) is a general-purpose model, not a specialized IAD method. The authors should clarify that this improvement is relative to general VLMs, not necessarily the state-of-the-art in IAD.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The manuscript addresses a relevant application domain (industrial anomaly detection) using agentic tools and reinforcement learning. From a safety and ethics perspective, the primary concerns revolve around transparency, reproducibility, and the potential for harm in deployment scenarios.

First, the **NeurIPS Paper Checklist** (Section **NeurIPS Paper Checklist**) contains a critical inconsistency regarding open science. Item 5 (“Open access to data and code”) is answered with “No,” justified by the promise to release data/code “upon the publication of our paper.” While common in preprints, this stance conflicts with the growing expectation for immediate reproducibility and open access in AI research. The authors should update the manuscript to specify the exact repository (e.g., GitHub, Zenodo), the license under which the code/data will be released, and a concrete timeline. Relying on a future release without a binding commitment or link reduces the paper’s scientific integrity and hinders independent verification of safety claims.

Second, the **methodology** relies on “Indus-CoT” generation via a “strong teacher model” (Section **sec:limitations**). The paper does not specify which model serves as the teacher or the provenance of the training data used to generate these Chain-of-Thought examples. In the context of industrial inspection, where false negatives can lead to physical defects or safety

hazards, it is crucial to disclose the source of the teacher model to assess potential biases. If the teacher model has known hallucination issues or biases in specific industrial domains, these could be amplified in the student model. The authors should add a brief disclosure in the “Limitations” or “Methodology” section identifying the teacher model and discussing any steps taken to mitigate bias in the generated CoT data.

Third, the **Safeguards** checklist item is marked “NA” with the justification that the method uses pre-trained models. This is insufficient for an agentic system that actively interacts with industrial data via tools (e.g., cropping, measuring). While the paper focuses on detection, the deployment of such an agent in a real-world setting carries risks. For instance, if the agent fails to detect a critical anomaly (false negative) due to tool limitations or model uncertainty, it could lead to the release of defective products or safety incidents. The authors should briefly address these deployment risks in the “Limitations” or “Broader Impacts” section and consider recommending a human-in-the-loop verification step for high-stakes applications.

Finally, the paper does not involve human subjects or crowdsourcing, so IRB/IACUC concerns are not applicable. However, the use of proprietary commercial models (e.g., GPT-4o, Claude 4) as baselines raises questions about the cost and accessibility of reproducing the results, which is an ethical consideration regarding the democratization of AI research. While not a fatal flaw, acknowledging this limitation would strengthen the paper’s ethical standing.

Overall, the paper is sound but requires minor revisions to address transparency in data/code availability, teacher model disclosure, and a more nuanced discussion of deployment risks.

Action items.

- **[writing]** The NeurIPS Checklist explicitly states ‘No’ for Open access to data and code, with justification that they will be released ‘upon publication.’ This creates a reproducibility barrier and potential conflict with open-science norms. Authors must clarify the specific license, release timeline, and repository URL in the final manuscript to ensure compliance with ethical data sharing standards.
- **[writing]** The paper relies on ‘Indus-CoT’ generated by a ‘strong teacher model’ (Section Limitations). The authors must explicitly disclose the identity of this teacher model, the data used to generate the CoT, and any potential biases introduced by the teacher’s training data to ensure transparency and prevent the propagation of hidden biases in the industrial inspection domain.
- **[writing]** The ‘Safeguards’ checklist item is marked ‘NA’ with the justification that the method uses pre-trained models. However, the system actively invokes tools (e.g., T_crop, T_measure) on industrial assets. Authors should briefly discuss potential risks if the agent is deployed in safety-critical environments (e.g., false negatives leading to undetected defects) and whether any human-in-the-loop verification is recommended.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_scientific_evidence.md

The paper presents IndusAgent, an agentic framework for open-vocabulary industrial anomaly detection, claiming state-of-the-art performance across five benchmarks. While the reported average scores (83.4%) and recall improvements (+17.4% on MPDD) are substantial, the scientific evidence supporting the robustness of these claims is currently insufficient due to a lack of statistical rigor and potential methodological ambiguities in the reinforcement learning setup.

First, the manuscript explicitly opts out of reporting statistical significance in the NeurIPS Checklist (Item 7), citing field norms. However, the ablation studies (Section 4.3, Tables 4 and 5) demonstrate massive performance variances when components are removed (e.g., VisA performance drops from 76.8% to 55.5% without SFT). In deep learning, particularly with RL and LLMs, results can be highly sensitive to random seeds, hyperparameters, and data shuffling. Without reporting standard deviations over multiple runs (e.g., 3-5 seeds) or confidence intervals, it is impossible to determine if the observed gains are statistically significant or artifacts of a specific random initialization. The claim of “catastrophic collapse” without SFT is dramatic but requires statistical backing to be scientifically rigorous.

Second, the reward formulation in Section 3.2 (Eq. 3) introduces a potential source of data leakage. The reward components R_{loc} (IoU) and R_{type} (semantic distance) require ground-truth annotations. The paper does not clarify whether the RL training process utilized the test set annotations to compute these rewards. If the agent was optimized to maximize metrics calculated on the test set, the reported SOTA results are invalid. Even if a validation set was used, the authors must explicitly state the separation of data splits for SFT, RL training, and final evaluation to ensure the integrity of the benchmark results.

Finally, the “Tool Usage Statistics” in the Appendix report a >98% success rate for tool calls. The definition of “success” is not provided. Does this mean the tool executed without crashing, or that the tool returned a visually useful crop? If the metric is merely API availability, it does not support the claim of “reliable” diagnostic reasoning. The evidence for the agent’s cost-awareness (penalizing redundant calls) is also weak without a breakdown of inference latency or computational cost per image compared to baselines.

To strengthen the scientific evidence, the authors must: (1) Report mean and standard deviation for all main results and ablation studies across multiple seeds; (2) Explicitly confirm that test set annotations were never used during the RL reward calculation; and (3) Define “tool success” and provide latency/cost metrics to substantiate the efficiency claims.

Action items.

- **[science]** The NeurIPS Checklist explicitly states ‘No’ for statistical significance (Item 7), claiming error bars are not required in the field. However, the ablation studies (Tables 4 & 5) show large performance swings (e.g., VisA dropping from 76.8% to 55.5% without SFT). The authors must provide standard deviations or results from multiple random seeds to confirm these gains are not due to random initialization or data split variance.
- **[science]** The reward function (Eq. 3) relies on ground-truth IoU (R_{loc}) and semantic distance (R_{type}) during training. The paper does not specify if the test set annotations used for these

rewards are the same as those used for evaluation, or if a separate validation set was used for RL tuning. If the RL policy was optimized directly on the test set metrics, this constitutes data leakage and invalidates the SOTA claim.

- **[science]** The ‘Tool Usage Statistics’ table (Appendix) reports a 99.1% success rate for tool calls. The definition of ‘success’ is ambiguous (e.g., API return 200 vs. correct visual crop). Without a clear definition and error analysis of failed tool calls, the claim that the agent is ‘cost-aware’ and ‘reliable’ is not fully supported by the evidence.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_statistical_analysis.md

The manuscript presents a novel agentic framework for industrial anomaly detection but lacks essential statistical rigor required to substantiate its claims of state-of-the-art performance. While the authors provide extensive point estimates in Tables 1, 2, 4, and 5, they fail to report any measure of uncertainty, such as standard deviations, confidence intervals, or p-values.

Specifically, in the “NeurIPS Paper Checklist” (Section e002), the authors explicitly answer “No” to the requirement for “Experimental result statistical significance,” justifying this by stating that the field “doesn’t require error bars.” This justification is insufficient for a paper claiming significant performance improvements (e.g., a 17.4% increase in Anomaly Recall on MPDD). In machine learning research, particularly with stochastic training procedures like Reinforcement Learning (GRPO) and Supervised Fine-Tuning, single-run results are prone to variance based on random seeds and data splits. Without reporting results averaged over multiple independent runs (e.g., 3-5 seeds) with standard deviations, it is impossible to determine if the reported gains are statistically significant or merely artifacts of a favorable random initialization.

Furthermore, the ablation studies in Section 4.3 (Tables 4 and 5) rely entirely on single-point comparisons. The dramatic performance drops observed when removing components (e.g., the drop to 55.5% on VisA without SFT) are compelling but lack statistical validation. The authors should re-run their experiments with multiple seeds to provide mean $\pm$ std. dev. metrics. Additionally, they should perform appropriate statistical significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) when comparing their method against baselines to rigorously support their claims of superiority. The current presentation of data, while numerically impressive, does not meet the standard of statistical evidence required for a robust scientific conclusion.

Action items.

- **[science]** The NeurIPS Checklist explicitly states ‘No’ for statistical significance (Item 7), justifying this by claiming the field ‘doesn’t require error bars.’ This is scientifically unsound for a paper claiming SOTA performance with specific margins (e.g., +17.4% on MPDD). The authors must report standard deviations or confidence intervals derived from multiple

random seeds or cross-validation folds to validate that these gains are not due to random initialization or data split variance.

- **[science]** The ablation studies (Tables 4 and 5) present single-point performance metrics without any measure of variance. Given the stochastic nature of RL training (GRPO) and SFT, the reported improvements (e.g., VisA dropping from 76.8% to 55.5% without SFT) need to be contextualized with statistical significance tests (e.g., paired t-tests or bootstrap confidence intervals) to ensure the observed effects are robust and reproducible.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript demonstrates a generally high standard of technical writing, with clear logical flow in the methodology and results sections. The introduction effectively sets the stage for the problem of open-vocabulary industrial anomaly detection. However, several specific areas require attention to polish the prose and eliminate minor grammatical errors that detract from the professional presentation.

First, the “NeurIPS Paper Checklist” in the supplementary material (e002) contains noticeable grammatical errors. In the justification for “Broader impacts,” the text reads “The border impacts is provided,” which should be corrected to “The broader impacts are provided.” Similarly, under “Safeguards,” the phrase “This proposed methods is safe” contains a subject-verb agreement error and should be “This proposed method is safe.” While these are in the checklist, they reflect on the overall attention to detail.

Second, the figure captions in the appendix (e001) are somewhat repetitive and could be more descriptive. The captions for `fig/case3.png` and `fig/case2.png` both state, “Case Study between Qwen3-VL-8B and our method.” This phrasing is slightly awkward and lacks specificity. It would be more effective to briefly describe the specific scenario or defect type being illustrated in each case study to guide the reader’s interpretation of the visual evidence.

Third, in the “Agentic Reinforcement Learning” section (e002), the transition to the GRPO formulation is slightly abrupt. The sentence “We utilize GRPO... to avoid actor-critic memory costs” is grammatically correct but could be improved for flow by briefly contextualizing why memory costs are a critical constraint for this specific agentic framework (e.g., due to the length of tool-use trajectories).

Overall, the paper is well-structured and readable. Addressing these minor grammatical slips and refining the figure captions will elevate the writing quality to match the technical contribution.

Action items.

- **[writing]** In the NeurIPS Checklist (e002), item ‘Broader impacts’ contains a typo: ‘The

border impacts is provided...’ should be ‘The broader impacts are provided...’. Additionally, item ‘Safeguards’ has a grammatical error: ‘This proposed methods is safe’ should be ‘This proposed method is safe’.

- **[writing]** In Appendix e001, Figure captions for ‘fig/case3.png’ and ‘fig/case2.png’ are repetitive and slightly awkward: ‘Case Study between Qwen3-VL-8B and our method.’ Consider rephrasing to ‘Comparison of Qwen3-VL-8B and IndusAgent on [specific task/defect type]’ for better clarity and flow.
- **[writing]** In Section ‘Agentic Reinforcement Learning’ (e002), the sentence ‘We utilize GRPO... to avoid actor-critic memory costs’ is slightly abrupt. Consider adding a brief clause explaining *why* actor-critic memory costs are a specific concern in this context (e.g., ‘...which are prohibitive for long-horizon tool-use trajectories’).